

Mahmoud Alyosify

Specialised LLM & Generative AI Engineer | Fine-Tuning, RAG & Inference Optimisation | MSc Artificial Intelligence, Queen's University (Canada)

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🌐 [linkedin.com/in/mahmoudalyosify](https://www.linkedin.com/in/mahmoudalyosify) | 🏠 github.com/MahmoudAlyosify | 📁 Portfolio | 🤖 Hugging Face | 🏆 Kaggle

PROFESSIONAL SUMMARY

LLM & Generative AI Engineer specialising in **fine-tuning, retrieval-augmented generation and inference-time optimisation** of open-weight language models. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**; master's research delivers a **training-free decoding-time method** that reduces generated tokens across **Mistral, Llama-3.1 and Qwen2.5** while proving semantic preservation under a pre-registered non-inferiority protocol. Hands-on with **LoRA/QLoRA fine-tuning at 159K-record scale**, self-hosted inference, LLM evaluation (BERTScore, ROUGE, LLM-as-Judge) and AWS GPU deployment. Focused on making language-model systems **cheaper, faster and measurably reliable** in production.

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Nov 2026
Expected Nov 2026
Queen's University, Kingston, Ontario, Canada

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Cloud Computing, End-to-End ML Pipelines.**

B.Sc. Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
GPA: 3.53 / 4.00
Assiut University, Egypt

- Graduation Project: **Vitalism Solution** — **Grade A+; Semi-Finalist, Microsoft Imagine Cup 2023.**

TECHNICAL SKILLS

LLM & Generative AI	Large Language Models, Instruction Fine-Tuning, LoRA, QLoRA, PEFT, RAG, Custom Decoding Strategies, LogitsProcessors, Sampling & Entropy-Based Gating, Prompt Engineering, Structured Outputs, Tool / Function Calling, Multi-Agent Systems
Frameworks & Serving	PyTorch, Hugging Face Transformers, PEFT, Datasets, Accelerate, LangChain, FAISS, ChromaDB, ONNX, FastAPI, Streamlit, Self-Hosted Inference, Docker, AWS GPU Instances
Models Worked With	Mistral-7B-Instruct, Llama-3.1-8B-Instruct, Meta-Llama-3-8B, Qwen2.5-3B-Instruct
LLM Evaluation	MT-Bench, XSum, SQuAD-v2, BERTScore, ROUGE, Exact-Match / F1, Length-Controlled LLM-as-Judge, Non-Inferiority Testing, Ablation Studies, Deterministic & Reproducible Benchmarking
Efficiency & Systems	Token-Cost Reduction, TTFT & Inter-Token Latency Profiling, GPU Energy Measurement (NVML), KV-Cache-Free Methods, Quantised Fine-Tuning (4-bit)
Programming	Python, SQL, C++, C#, JavaScript, Bash, Git/GitHub, Linux

SELECTED LLM RESEARCH & ENGINEERING PROJECTS

Verbosity-Aware Decoding for Token-Efficient Language Model Inference 2026 – Present
MSc Research Project (CISC 898), Queen's University · Supervisor: Dr. Rahatara Ferdousi

- Designed a **training-free LogitsProcessor** that reduces generated tokens at **decoding time** by fusing stop-token, n-gram-repetition and semantic-redundancy signals under an **entropy-driven gate** — no retraining, no architecture change, no KV-cache or attention overhead, and drop-in compatible with Hugging Face generate.
- Validated across **three open-weight model families** (Mistral-7B-Instruct, Llama-3.1-8B-Instruct, Qwen2.5-3B-Instruct) on **MT-Bench, YapBench, XSum** and **SQuAD-v2**, covering open-ended chat, summarisation and extractive QA.
- Built a **pre-registered evaluation protocol** that proves semantic preservation via paired **non-inferiority testing** (BERTScore, ROUGE, exact-match/F1, length-controlled LLM judge) alongside token reduction, **TTFT / inter-token latency** and **GPU energy** through NVML — fully deterministic and reproducible on pinned AWS GPU instances.

Horus-OSINT — Domain-Adapted LLM Assistant on AWS 2026
Meta-Llama-3-8B · Instruction Fine-Tuning · AWS · Python

- Fine-tuned **Meta-Llama-3-8B** on **159,826** curated GTD/GDELT records, building the full pipeline from dataset construction and instruction formatting through training to evaluation.
- Deployed the adapted model on **AWS** as an interactive assistant that answers natural-language questions over global event and threat-intelligence data.

HORUS Sentinel — Multi-Agent LLM Platform over a Knowledge Graph 2026
Multi-Agent Systems · Fine-Tuned LLM · Knowledge Graphs · Self-Hosted Inference

- Architected a swarm of specialised agents that continuously collect open-source data and correlate every finding into a **living knowledge graph** used as the retrieval substrate for generation.
- A **self-hosted, fine-tuned language model** reasons over that graph to produce **prioritised, evidence-backed reports**, grounding

every claim in retrieved source material.

Automated Multimodal Agent — PDF to Narrated Presentation
Generative AI · Agentic Orchestration · Multimodal Generation

2026

- Built an autonomous think–plan–act workflow that converts PDFs into fully narrated PowerPoint decks by chaining document understanding, structured slide generation and audio synthesis.

Additional Relevant Work

- **Human vs. Agentic Pull Requests** — reproducible GitHub-mining study quantifying review effort and failure modes of LLM-generated versus human-authored code.
- **SimCLR-Vision-SSL** — contrastive representation learning with **ONNX/FAISS** embedding retrieval — transferable to RAG vector-search pipelines.

PROFESSIONAL EXPERIENCE

Machine Learning Instructor <i>National Telecommunication Institute (NTI)</i>	<i>Jul 2025 – Sep 2025</i> Cairo, Egypt
• Delivered a 90-hour hands-on ML program covering supervised learning, PCA and neural networks with practical labs throughout.	
AI in Cybersecurity Intern <i>e& Egypt — via NTI (On-the-Job Training)</i>	<i>Jun 2025 – Jul 2025</i> Cairo, Egypt
• Applied AI-driven tooling across threat intelligence, digital forensics and enterprise security analysis.	
Systems Engineer (Military Service) <i>National Company for Roads Building & Development</i>	<i>Feb 2024 – Mar 2025</i> Cairo, Egypt
• Supported backend systems for national toll infrastructure serving 50%+ of Egyptian vehicles ; co-developed an internal data-management application in C#, .NET, SQL Server .	
Online Instructor — AI, Algorithms & Data Science <i>Udemy & YouTube</i>	<i>2020 – Present</i> Remote
• Built Arabic-language technical curricula reaching 10,000+ learners and 700K+ YouTube views .	

CERTIFICATIONS & AWARDS

Generative AI & Prompt Engineering — IBM / Coursera	AWS Academy — ML for NLP, ML Foundations, Data Engineering
Machine Learning Specialization — DeepLearning.AI	HCIP – AI (Huawei)
Digilians Presidential Scholarship — Queen's University	Semi-Finalist — Microsoft Imagine Cup 2023
1st Place — Smart Cities Hackathon 2022	Ideal Student Award — Assiut University, 2022–2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.
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